

The Master Recursion for $Q_{n,c}(q)$, the Preimage EMD Dichotomy, and the Shape of the Level-2 Numerator

Seed 7, Layer 2, Round 2

2026-07-04

Abstract

Composing the (proved) adjugate inversion of the Corteel–Welsh system with the conversion $Q_{n,c} = (q^\ell; q^\ell)_n g_{c,n}$ yields an explicit one-step recursion expressing Q_n in terms of Q_{n-1} and Q_{n-2} via Earth-Mover’s-Distance weights on profile space (Theorem ??). We prove a dichotomy for the behaviour of $\text{EMD}(c, \cdot)$ under the three 2-shift preimage moves (Lemma ??) and deduce that all negativity in the level-2 numerator $N_2 = (1 + q^2 + q^4)Q_2$ is confined to three rigid “sandwich” shapes with a single negative coefficient each (Theorem ??). We also show that divisibility of N_n by $1 + q^n + q^{2n}$ factors through a single scalar identity per root, generalizing the level-1 EMD Equidistribution Theorem. Positivity of N_n (verified for $n \leq 5$ at $d = 4$ and $n \leq 2$ at $d \in \{5, 7\}$) remains conjectural.

1 Setup and notation

Fix $r = 3$ and a profile $c = (c_0, c_1, c_2)$ with $d = c_0 + c_1 + c_2$, $\ell = \gcd(d, 3)$. Write $G_c(z, q) = (zq; q)_\infty F_c(z, q)$, $g_{c,n} = [z^n]G_c(z, q)$ and $Q_{n,c} = (q^\ell; q^\ell)_n g_{c,n}$. For $J \subseteq I_c = \{i : c_i > 0\}$ let $c(J)$ be the Corteel–Welsh shifted profile. The Earth Mover’s Distance on profiles satisfies (proved in Round 2 Layer 1, Seed 4/Seed 7)

$$\text{EMD}(c, c') = 3 \max(0, c'_1 - c_1, c_0 - c'_0) + (c'_0 - c_0) - (c'_1 - c_1), \quad (1)$$

$\text{adj}(I - A(x))[c, c'] = x^{\text{EMD}(c, c')}$ and $\det(I - A(x)) = -(x^3 - 1)$, whence the *adjugate inversion*

$$g_{c,n} = \frac{1}{1 - q^{3n}} \sum_{c'} q^{n \cdot \text{EMD}(c, c')} b_n(c'). \quad (2)$$

2 The master recursion

Theorem 1 (Master recursion). *For $n \geq 1$ (with $Q_{-1} := 0$, $Q_{0,c} := 1$),*

$$Q_{n,c} = \frac{1 - q^{\ell n}}{1 - q^{3n}} \sum_{c'} q^{n \text{ EMD}(c,c')} \text{BR}_n(c'),$$

where

$$\begin{aligned} \text{BR}_n(c') = & q^{2n-1} \sum_{\substack{J \subseteq I_{c'} \\ |J|=2}} Q_{n-1,c'(J)} + [\text{rank}(c') = 3] \left(-(q^{3n-2} + q^{3n-1}) Q_{n-1,c'} \right. \\ & \left. + q^{3n-3} (1 - q^{\ell(n-1)}) Q_{n-2,c'} \right). \end{aligned}$$

For $\ell = 1$ the prefactor is $1/(1 + q^n + q^{2n})$.

Proof. Extract $[z^n]$ from the Corteel–Welsh equation for G_c , $G_c(z, q) = \sum_{\emptyset \neq J \subseteq I_c} (-1)^{|J|-1} (zq; q)_{|J|-1} G_{c(J)}(zq^{|J|}, q)$, using $(zq; q)_0 = 1$, $(zq; q)_1 = 1 - zq$, $(zq; q)_2 = 1 - z(q + q^2) + z^2 q^3$. The z^0 -coefficients of the truncated Pochhammers produce the matrix $A(q^n)$ acting on the current level; the remaining terms give

$$b_n(c) = q^{2n-1} \sum_{\substack{J \subseteq I_c \\ |J|=2}} g_{c(J),n-1} - [\text{rank}(c) = 3] (q^{3n-2} + q^{3n-1}) g_{c,n-1} + [\text{rank}(c) = 3] q^{3n-3} g_{c,n-2},$$

where we used that $|J| = 3$ forces $c(J) = c$ and requires $\text{rank}(c) = 3$. Insert this into (??) and substitute $g_{c,m} = Q_{m,\cdot}/(q^\ell; q^\ell)_m$, multiplying through by $(q^\ell; q^\ell)_n$ and using $(q^\ell; q^\ell)_n/(q^\ell; q^\ell)_{n-1} = 1 - q^{\ell n}$ and $(q^\ell; q^\ell)_n/(q^\ell; q^\ell)_{n-2} = (1 - q^{\ell n})(1 - q^{\ell(n-1)})$. At $n = 1$ the $g_{c,-1}$ term is absent. $\square$

Remark 2. *Verified against an independent Neumann-iteration solve of the CW linear system (no adjugate used): $d = 4$, $n \leq 3$; $d = 5, 7$, $n \leq 2$; all profiles, exact coefficient match, and $Q_{2,c}(1) = \left(\frac{(d+1)(d+2)}{6} - 1\right)^2$. The fully unfolded $n = 2$ double sum, obtained by inserting the closed form $Q_{1,c} = \frac{1}{1+q+q^2} \sum_{c'} q^{\text{EMD}(c,c')} B(c')$ ($B = q(2-q)$, $q, 0$ for ranks 3, 2, 1), also matches exactly.*

3 The preimage EMD dichotomy

For a profile c' call c'' a 2-shift preimage of c' if $c''(J) = c'$ for some $J \subseteq I_{c''}$ with $|J| = 2$. Explicitly the three candidate moves are

$$J = \{0, 1\} : c'' = c' + e_0 - e_2, \quad J = \{1, 2\} : c'' = c' - e_0 + e_1, \quad J = \{0, 2\} : c'' = c' - e_1 + e_2.$$

A direct support check shows: a rank-3 profile c' has exactly three preimages, a rank-2 profile exactly one, a rank-1 profile none.

Lemma 3 (Preimage EMD dichotomy). *Fix c and c' , put $e_0 = \text{EMD}(c, c')$, and for each existing preimage c'' put $\Delta = \text{EMD}(c, c'') - e_0$. Then $\Delta \in \{-2, +1\}$, and at most one of the three preimages has $\Delta = -2$.*

Proof. Set $u = c'_0 - c_0$, $v = c'_1 - c_1$, so by (??) $e_0 = f(u, v) := 3M(u, v) + u - v$ with $M(u, v) = \max(0, v, -u)$. The three preimage moves act on (u, v) by $a = (+1, 0)$, $b = (-1, +1)$, $c = (0, -1)$.

Move a: the linear part $u - v$ increases by 1; $M(u+1, v) = \max(0, v, -u-1) \in \{M, M-1\}$. Hence $\Delta_a \in \{1, -2\}$, and $\Delta_a = -2$ iff $M = -u$ and $-u-1 \geq \max(0, v)$, i.e. $-u \geq \max(0, v) + 1$ (in particular $u \leq -1$).

Move c: the linear part increases by 1; $M(u, v-1) \in \{M, M-1\}$, and $\Delta_c = -2$ iff $v \geq \max(0, -u) + 1$ (in particular $v \geq 1$).

Move b: the linear part decreases by 2; $M(u-1, v+1) = \max(0, v+1, -u+1) \in \{M, M+1\}$, so $\Delta_b \in \{-2, 1\}$, and $\Delta_b = -2$ iff $M(u-1, v+1) = M$, which forces $M = 0$ with $v+1 \leq 0$ and $-u+1 \leq 0$, i.e. $v \leq -1$ and $u \geq 1$.

Mutual exclusivity of the three conditions: $\Delta_a = -2$ needs $u \leq -1$ while $\Delta_b = -2$ needs $u \geq 1$; $\Delta_c = -2$ needs $v \geq 1$ while $\Delta_b = -2$ needs $v \leq -1$; and $\Delta_a = -2$ with $\Delta_c = -2$ gives $-u \geq v+1$ and $v \geq -u+1$, whence $-u \geq -u+2$, absurd. $\square$

(Computationally re-checked for $d \in \{4, 5, 7, 8, 10\}$, all pairs (c, c') : zero violations; both values $\{-2, 1\}$ occur for ranks 2 and 3.)

4 The shape of the level-2 numerator

Let $\ell = 1$ and define $N_2(c) := (1 + q^2 + q^4) Q_{2,c} = \sum_{c'} q^{2\text{EMD}(c, c')} \text{BR}_2(c')$. Regrouping the sum by the profile appearing as the argument of Q_1 and applying Lemma ?? gives:

Theorem 4 (Shape theorem). *Fix c and write $e_0 = e_0(c') = \text{EMD}(c, c')$, and for rank-3 profiles let $k(c, c') \in \{0, 1\}$ be the number of preimages of c'*

with $\Delta = -2$. Then

$$\begin{aligned}
N_2(c) = & \sum_{\text{rank}(c')=2} q^{2\text{EMD}(c,c')+3} Q_{1,c'} \quad (c'' = \text{the unique preimage of } c') \\
& + \sum_{\substack{\text{rank}(c')=3 \\ k=0}} q^{2e_0+4} (2q-1) Q_{1,c'} + \sum_{\substack{\text{rank}(c')=3 \\ k=1}} q^{2e_0-1} (1-q^5+q^6) Q_{1,c'} \\
& + \sum_{\text{rank}(c')=3} q^{2e_0+3} (1-q).
\end{aligned}$$

Proof. By Theorem ?? at $n = 2$, $N_2(c) = \sum_{c'} W(c, c') Q_{1,c'} + \sum_{\text{rank}(c')=3} q^{2e_0+3} (1-q)$ with $W(c, c') = q^3 \sum_{c''} q^{2\text{EMD}(c,c'')} - [\text{rank}(c') = 3](q^4 + q^5)q^{2e_0}$, the inner sum over preimages c'' of c' ; the last term is the Q_0 contribution $q^3(1-q)Q_{0,c'} = q^3(1-q)$. For rank-2 profiles W is the single monomial shown. For rank-3 profiles all three preimages exist and by Lemma ?? their exponents are $2(e_0 + \Delta_i) + 3$ with $\Delta_i \in \{-2, 1\}$ and at most one -2 : if $k = 0$, $W = 3q^{2e_0+5} - q^{2e_0+4} - q^{2e_0+5} = q^{2e_0+4}(2q-1)$; if $k = 1$, $W = q^{2e_0-1} + 2q^{2e_0+5} - q^{2e_0+4} - q^{2e_0+5} = q^{2e_0-1}(1-q^5+q^6)$. $\square$

Corollary 5. *Every negative coefficient of any single term of $N_2(c)$ arises from one of the three sandwich polynomials $2q-1$, $1-q^5+q^6$, $1-q$, each contributing exactly one negative coefficient per rank-3 profile c' (namely $-q^{2e_0+4}(Q_{1,c'}+1)$ in total).*

5 Equidistribution at level n

Theorem 6 (Scalar reduction of divisibility). *Let $\gcd(d, 3) = 1$ and ζ be any root of $1 + q^n + q^{2n}$, so that $\omega_\zeta := \zeta^n$ is a primitive cube root of unity. Since $\text{EMD}(c, c') \equiv \delta(c') - \delta(c) \pmod{3}$ with $\delta(c) = (c_0 - c_1) \pmod{3}$,*

$$N_n(c)|_{q=\zeta} = \omega_\zeta^{-\delta(c)} S_n(\zeta), \quad S_n(\zeta) := \sum_{c'} \omega_\zeta^{\delta(c')} \text{BR}_n(c')|_{q=\zeta}.$$

Hence divisibility of all $|\mathcal{P}_d|$ numerators $N_n(c)$ by $1 + q^n + q^{2n}$ is equivalent to the single scalar identity $S_n(\zeta) = 0$ per root ζ .

Proof. Immediate from $\zeta^{n\text{EMD}} = \omega_\zeta^{\text{EMD}}$ and the mod-3 formula for EMD (Layer 1, Seed 7). The identity $S_n(\zeta) = 0$ itself follows from Welsh's polynomiality of Q_n ; at $n = 2$ it was also verified directly ($S_2(\omega) = S_2(-\omega) = 0$ for $d = 4, 5, 7$). This is the level- n generalization of the EMD Equidistribution Theorem: the c -dependence factors out completely, leaving one global charge-conservation identity. $\square$

6 What remains open

Conjecture 7 (Numerator positivity). $N_n(c) = (1 + q^n + q^{2n}) Q_{n,c} \geq 0$ coefficientwise for all n, c (when $\gcd(d, 3) = 1$).

Verified: $d = 4, n \leq 5$, all 15 profiles (exact arithmetic); $d = 5, 7, n \leq 2$, all profiles. Two further facts constrain any proof: (i) $(1+q+q^2)Q_2 \geq 0$ holds but $(1 - q + q^2)Q_2 \geq 0$ *fails* for every profile tested, so positivity descends from N_2 through the Φ_6 division first and the Φ_3 division last; (ii) the profile monotonicity $Q_{1,c'(J)} \geq q^a Q_{1,c'}$ is false for all $a \geq 1$, so no termwise pairing inside Theorem ?? can work: positivity of N_2 is an irreducibly global, EMD-weighted phenomenon. By Theorem ?? the required cancellation is now confined to a single negative coefficient per rank-3 profile, a drastic reduction of the search space for a sign-reversing injection.